

REPUBLIC OF CAMEROON

Peace – Work – Fatherland

MINISTRY OF SECONDARY EDUCATION

INSPECTORATE GENERAL OF EDUCATION

Inspectorate of Pedagogy in charge of the
Teaching of Computer Science



REPUBLIQUE DU CAMEROUN

Paix – Travail – Patrie

MINISTRE DES ENSEIGNEMENTS SECONDAIRES

INSPECTION GENERALE DES ENSEIGNEMENTS

Inspection de Pédagogie chargée de
l'enseignement de l'informatique

NATIONAL HARMONISED PROGRESSION SHEET FOR COMPUTER SCIENCE FORM 4

SCHOOL YEAR.....

SCHOOL.....

WEEKLY WORKLOAD: 3 periods

TEACHER.....

GRADE.....

TEL.....

Term	Week	Module	Category of action	Competency statement	Lesson title	Objectives	Resources	Nature of lesson		
								Th	Prac	Dig
FIRST TERM	1	Hardware and software systems 1	Creating digital content 1 (using AI)	Faced with situation related to creating digital content, learners propose and use appropriate too	Diagnostic evaluation Lesson 1: Introduction to Artificial Intelligence	- Define AI - Explain the history and evolution of AI - Identify and categorise the different types of AI in real world				
					Lesson 2: Applications and Ethics of Artificial Intelligence	- Identify common applications of AI - Explain the uses of AI in different fields - State the ethical issues related to AI				
					Lesson 3: The use of appropriate Prompts to generate AI responses	- Examine prompt results - Discuss prompt results - Write appropriate prompts to solve problems within a given context.				
					Lesson 4: Integration activities					
	2	Problem solving and coding 2	Using control structures	Given an algorithmic problem that requires the use of control structures, learners produce algorithms that use the appropriate control structure to solve the problem.	Lesson 5: sequence and selection control structure	- Explain how sequence and selection control structures work. - Illustrate the functioning of sequence and selection control structures with a flowchart. - Write simple algorithms that make use of a selection control structure.				
					Lesson 6: Multiple selection constructs	- Explain how the different types of multiple selection constructs work - Illustrate the functioning of the different multiple selection constructs - Write algorithms that make use of multiple				

	3					selection control structures				
					Lesson 7: Iterative constructs	- Identify situations where an iterative or loop control structure is needed. - Describe the 4 main parts of a loop. - Explain advantages and disadvantages of iterative control structures.				
					Lesson 8: Definite iterative constructs	- Choose correctly when to use a definite iteration. - Build definite iterations that are coherent with a given context. - Write simple algorithms that make use of definite iteration				
					Lesson 9: Indefinite iterative constructs	- Differentiate between pretest and post-test indefinite iterations. - Illustrate the functioning of indefinite iterations using a flowchart. - Write simple algorithms that make use of indefinite iteration.				
					Lesson 10: Flowchart representation	Represent algorithms that make use of control structures as a flowchart.				
					Lesson 11: Integration activities					
	4	Testing and debugging	Given an algorithm, learners evaluate its semantic correctness using the dry-run technique and propose a way of fixing errors if any.	Lesson12: Notions on testing and debugging	- Explain the concepts of testing, test case, test data and bug. - Differentiate between testing and debugging. - Differentiate between black box test and white box test.					
				Lesson 13: Dry running	- Explain the concepts of dry running and trace tables. - Perform a dry run test on an algorithm.					
				Lesson 14: Integration activities						
	5	Ethics, Society and legal issues 2	Practising netiquette rules	Placed in a context that requires communicating with others, learners determine and apply appropriate behavioural norms that respect others and their cultural and generational diversities.	Lesson 15: Notions on netiquette	- Explain the concepts of netiquette, and empathy. - State codes of conduct for posting and sharing content online. - Explain the importance of empathy, awareness in cultural and generational diversities when communicating online.				
					Evaluation					
					Remediation					
					Lesson 16: Emojis	- Differentiate emojis from emoticon. - Give the meaning of common emojis. - Explain why it is important to know the meaning of an emoji before using it.				
					Lesson 17: Communicating responsibly online	Choose communication modes and strategies adapted to an audience.				
	6									
7										

						- Identify hostile or derogatory messages or activities online that attack an individual or groups of individuals. - Propose behavioural rules when using digital technologies in a context.						
					Lesson 18: Integration activities							
		8	Operations on number systems	Presented with a situation that involves number systems, learners apply appropriate operations on number systems to solve the problem.	Lesson 19: Notions on number systems	- Outline symbols of base 2, 8, 10, and 16. - Identify symbols in a given base - Recognise a number in a given number system						
	Lesson 20: Conversion from any base, to base 10				- Explain the principle of conversion from any base to base 10. - Convert from base 2, 8, and 16 to base 10.							
	Lesson 21: Conversion from base 10, to any base				- Explain the principle of conversion from base 10 to any base - Convert from base 10 to base 2, 8, and 16							
	Lesson 22: Conversion from base 2, to 8 and 16				- Explain the principle of conversion from base 2 directly to base 8, and 16 - Convert from base 2, to base 8, and 16. - Explain the principle of conversion from base 8 or 16 directly to base 2							
	Lesson 23: Conversion from base 16, to base 2 and 8, and addition in base 2, 8 and 16				- Convert from base 16, to base 2 and 8 directly. - Explain the principle of addition in any number system - Apply the principle of addition to add in a base 2, 8, and 16.							
	Lesson 24: Subtraction in a number system				- Explain the principle of subtraction in any number system - Apply the principle of addition and subtraction to add and subtract in a base 2, 8, and 16.							
	Lesson 25: Integration activities											
	9				Analysing simple logic circuits	Learners analyse a situation concerning logic circuits and determine the correct outputs and truth tables of the different elements used in the circuit.	Lesson 26: Basic logic gates (NOT, AND, OR)	- Explain the concept of logic gates and truth tables. - Identify a given basic logic gate. - Explain the functioning of a given basic logic gate. Produce the truth table of a given basic logic gate.				
							Lesson 27: Derived gates (NAND, NOR, XOR)	- Identify a given logic gate. - Explain the functioning of a given logic gate. Produce the truth table of a given logic gate.				
							Lesson 28: Analyse Logic circuits	- Explain the concept of logic circuit. - Deduce the logic expression of a logic circuit. Deduce the output value of logic expression or logic circuit when given specific input values.				
							Lesson 29: Draw Logic circuits	Produce the logic circuit of a given logic expression. Deduce the output value of logic expression or logic				
	10											

SECOND TERM	12	Ethics, Society and legal issues 2	Evaluating the credibility and reliability of information	Given a problem with factors related to the reliability and credibility of information, learners propose strategies of verifying the information. The strategies proposed should be pertinent, coherent, and logical.		circuit when given specific input values.				
					Lesson 30: Integration activities					
					Evaluation					
					Remediation					
	13	Ethics, Society and legal issues 2	Evaluating the credibility and reliability of information	Given a problem with factors related to the reliability and credibility of information, learners propose strategies of verifying the information. The strategies proposed should be pertinent, coherent, and logical.	Lesson 31: Notions on misinformation and disinformation	• Explain the concepts of misinformation, disinformation, deepfake, infodemic, clickbait. • Differentiate between misinformation and disinformation. • State factors of information biases (data algorithms, censorship, editorial choices, personal limitations).				
					Lesson 32: Analysis of online information	• Identify the author and source of a given information found online. • Identify sponsored content online. • Analyse information to detect its purpose or interest.				
					Lesson 33: Identifying the credibility of information	• Explain strategies for verifying the credibility and reliability of information. • Propose national and international sources of getting credible information.				
					Lesson 34: Integration activities					
	14	Hardware and software systems 2	Creating digital content 2	Faced with a situation related to the creation and editing of digital content, learners propose a variety of tools that can be used to solve the problem and produce digital content using an appropriate tool.	Lesson 35: Types of Digital content and File formats	• Describe the main types of digital content and their commonly used file formats. • Identify appropriate traditional and AI tools for producing different types of digital content. • Differentiate between multimedia and hypermedia				
					Lesson 36: Creation of audio and image content	• Produce audio content using hardware tools. • Create and edit audio content using traditional software and AI tools.				
					Lesson 37: Creation of image content	• Produce image content using hardware tools. • Produce and edit image content using traditional software and AI tools.				
					Lesson 38: Creation of multimedia content	• Produce multimedia content using hardware tools. • Produce and edit multimedia content using traditional software and AI tools.				
					Lesson 39: Creation of hypermedia content 1	• Identify the necessary technologies to create hypermedia content. • Create hypermedia content with the help of html, CSS, JavaScript, etc.				
					Lesson 40: Creation of hypermedia content 2	• Create hypermedia content with the help of html, CSS, JavaScript, etc.				
	15				Lesson 41: Integration activities					

					Evaluation					
					Remediation					
16	Problem solving and coding	Setting up a programming environment	Given a problem, learners select the appropriate tools for programming and install them.	Lesson 42: Programming tools	- State the role of the following programming tools: text editor, translator, and IDE. - Differentiate between a compiler and an interpreter. - Explain the advantages and disadvantages of installing an IDE over separate programming tools and vice-versa.					
				Lesson 43: Installation of programming tools	- Install an IDE and use it to run a program - Test the functionalities of an installed programming tool.					
Writing of source code		Given an algorithm, learners transform it into a syntactically and semantically correct program.	Lesson 44: Introduction to coding (Python)	- Explain the concept of coding. - Explain strategies that can be used to ease coding. - Recognise the structure of a program in a programming language and write source code that makes use of input, output, mathematical, and assignment operators.						
			Lesson 45: Coding 1 (Python)	- Write source codes that makes use of selection control structures. - Write source codes that make use of multiple selection control structures.						
			Lesson 46: Coding 2 (Python)	Write source codes that make use of definite iterative control structures						
			Lesson 47: Coding 3 (Python)	Write source code that make use of indefinite iterative control structures						
			Lesson 48: Integration activities							
17			Calculating storage space	Given a problem with factors related to storage space and units, learners determine the cause and solution to the problem showing clearly how they achieved the proposed conclusions.	Lesson 49: Units of storage and conversion	- Explain the read and write operations on storage. - State units of storage and the relationship between these units. - Explain how to convert from a given unit to another and convert between units of storage.				
					Lesson 50: Conversion between units of storage	- Explain how to convert from a given unit to another. - Convert between units of storage.				
18				Lesson 51: Integration activities						
				Evaluation						
	Hardware and software			Remediation						
				Lesson 52: Features of a word processor	- Choose appropriate word processor features for a given task. - Perform basic formatting.					

	21		documents	produce or reproduce content, learners use appropriate features of a word processor to achieve the task.	Lesson 53: Common operations on a page	• Explain what is meant by page layout • Perform basic operations on a page				
					Lesson 54: Operations on tables	• Create tables of different rows and columns using a word processor. • Perform operations on tables (add, delete, resize, ...)				
					Lesson 55: Integration activities					
	22		Manipulating spreadsheets	Given tasks that are related to spreadsheets and with appropriate guidance, learners choose functions or spreadsheet features that are coherent with the tasks and apply them correctly on spreadsheet software.	Lesson 56: Notions on cell referencing	• Explain the concepts of cells, formula, ranges, and cell reference. • Differentiate between the different types of cell referencing. • Identify when to use a given type of cell referencing.				
					Lesson 57: Operations using different types of cell referencing	• Perform operations (sum, average, product, count) using different types of cell referencing.				
					Lesson 58: Conditional functions: IF function	• Describe the IF function. • Identify when to use the IF function. • Solve variety of problems using the IF function.				
					Lesson 59: Conditional functions: SUMIF, PRODUCTIF	• Describe the SUMIF, and PRODUCTIF functions. • Identify when to use the SUMIF, and PRODUCTIF function. • Solve a problem that uses the SUMIF, and PRODUCTIF.				
					Lesson 60: Conditional functions: COUNTIF, AVERAGEIF	• Describe the COUNTIF, AVERAGEIF functions. • Identify when to use the COUNTIF, AVERAGEIF function. • Solve a problem that uses the COUNTIF, AVERAGEIF.				
					Lesson 61: Represent data using charts	• Identify the different types of charts. • Represent data using charts. • Perform simple formatting on charts.				
					Lesson 62: Integration activities					
	24		Avoiding computer related health issues	Given a problem or a situation related to avoiding computer related issues, learners evaluate the workplace, practices and habits of users based on standards and report on elements to reinforce and	Lesson 63: Computer related health hazards	• Describe different types of computer related health hazards. • Outline major causes of computer related health hazards.				
					Lesson 64: Computer ergonomics: best practices in the design of items and workplace setup	• Identify wrong equipment placement and design in a computing environment. • Explain how the design of equipment and workplace setup can help prevent computer related health injuries,				
Lesson 65: Computer ergonomics: workplace habits and exercises		• Identify wrong user posture and habits in a computing environment.								
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				elements to adjust. The recommendations should express appropriate practices and health issues that can arise for not respecting these practices.		• Explain how the workplace habits and exercises can help prevent computer related health injuries.				
					Lesson 66: Integration activities					
					Evaluation					
	26				Remediation					
					Lesson 67: Software methods for optimising the computer	• Outline common software for optimising the performance of a computer. • Explain the following concepts and how they affect the performance of a computer: malware, fragmentation, defragmentation. • Explain how antivirus, disk defragmenter, disk cleaner, and operating system configurations can improve the performance of a computer.				
					Lesson 68: Hardware methods for optimising the computer	• Explain how size of RAM and disk can improve computer performance. • Explain how SSDs improve computer performance compared to HDDs. • Explain hardware related methods for improving computer performance.				
					Lesson 69: Integration activities					
	27				Lesson 70: Notions on computer maintenance	• Explain the concepts of maintenance, hardware maintenance, software maintenance, preventive maintenance, and corrective maintenance. • Classify maintenance problems into hardware and software. • Classify solutions into preventive and corrective maintenance.				
					Lesson 71: Hardware maintenance	• Match symptoms to a given hardware problem. • Determine appropriate and logical means of solving a given hardware problem. Explain common preventive maintenance tips.				
					Lesson 72: Software maintenance	• Match symptoms to a given software problem. • Determine appropriate and logical means of solving a given software problem. Explain common preventive maintenance tips.				
					Lesson 73: Integration activities					
	28				Lesson 74: Types of sensors	• Explain the concept of sensors, and actuators. • State the different types of sensors and their roles. Outline situations in life where a given type of sensor				

			and Cloud Computing	embedded systems, learners identify the correct type of sensors used in the situation and their role.		can be used.				
	29				Lesson 75: Embedded systems and IoT	Differentiate between embedded systems, IoT, and cloud computing. Describe scenarios where IoT and embedded systems are used in life.				
					Lesson 76: Cloud computing	- Explain the concept of cloud computing. - Outline the 3 main types of cloud services with examples. - Describe how embedded systems and IoT relate to cloud computing.				
					Lesson 77: Integration activities					
	30	Ethics, society and legal issues 2	Appraising National and International Laws on Cybersecurity	Learners state national and international laws or acts that are related to protecting and regulating activities in the cyberspace.	Lesson 78: International laws or acts on digital technology	- Explain the importance of laws and acts to regulate activities related to the creation and use of data and digital technologies. - Outline international laws and acts related to the use of digital technology. - State the purpose of a given international law or act.				
					Lesson 79: National laws or policies on digital technology	- State laws or policies aimed at regulating the use of digital technology in Cameroon and their respective goals. - Discuss some elements of law No 2010-12 of 21st December - State penalties of some information and communication technology malpractice in Cameroon.				
					Lesson 80: National and international bodies regulating the use of ICT	- Outline international bodies that regulate the use of ICT and their respective goals. - Outline national bodies in Cameroon that regulate the use of ICT and their respective goals.				
					Lesson 81: Integration activities					
	31	Hardware and software systems 2	Choosing appropriate peripheral devices	Learners select appropriate peripheral devices based on a situation or context. Clear justification of the choice should be made.	Lesson 82: Input peripherals	- State the purpose of a given input peripheral device - Describe the characteristics of a given input peripheral device - Choose appropriate input peripheral in a given context				
					Lesson 83: Output peripherals	- State the purpose of a given output peripheral device - Describe the characteristics of a given output peripheral device - Choose appropriate output peripheral in a given context				
	32				Lesson 84: Integration activities					

					Evaluation					
					Remediation					
	33	Revision								
	34									
	35									
	36	END OF PROGRAM								